Machine Learning in Medical Diagnostics

AI System for Early Recognition & Prediction of Chronic Diseases

An interactive medical diagnostic platform predicting Heart Disease, Parkinson's, Breast Cancer, and Diabetes with high precision.

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Tech Stack
Python • Streamlit • Scikit-Learn

ROADMAP

Presentation Agenda

01

Introduction & Goals

Significance of early detection and unified diagnostic vision.

02

AI & Machine Learning

Machine learning lifecycle and clinical classification algorithms.

03

Tech Stack & Tools

Python environment, Streamlit framework, and core ML libraries.

04

Execution & Challenges

System interface, feature parameters, and operational challenges.

PROJECT SCOPE

Targeted Chronic Diseases

Heart Disease

Leading cause of global mortality requiring high-precision early prediction.

Parkinson's Disease

Progressive nervous disorder identified through vocal acoustic measurements.

Breast Cancer

Highly prevalent cancer type where therapeutic outcome relies on early diagnosis.

Diabetes

Metabolic condition causing long-term physiological complications if undetected.

CORE OBJECTIVES

System Goals & Value Proposition

Rapid Screening

Drastically cuts diagnostic delay for immediate intervention.

High Accuracy

Optimized algorithms reducing diagnostic human error.

Unified Platform

Consolidates four distinct ML models into one web suite.

Cost Reduction

Minimizes invasive, late-stage emergency treatment expenses.

THEORETICAL FOUNDATION

Machine Learning Development Lifecycle

1

Data Collection

Gathering validated clinical patient datasets.

2

Preprocessing

Handling missing values and scaling features.

3

Model Building

Algorithm selection and model training.

4

Evaluation

Cross-validation and hyperparameter tuning.

5

Deployment

Integrating saved models into Streamlit.

TECHNICAL STACK

Software Environment & Libraries

Python Language

Core language leveraged for data processing and AI framework ecosystem.

Streamlit Framework

Enables rapid building of interactive Web UI for real-time model inference.

Scikit-Learn & Pickle

Algorithms classification suite and `.sav` serialization for model storage.

ARCHITECTURE

System Workflow & Data Pipeline

1. User inputs patient health metrics into the interactive Streamlit UI.
2. System loads pre-trained serialized models via Pickle (.sav files).
3. Real-time feature processing & prediction in Python backend environment.
4. Instant diagnostic output display with clinical decision support options.

Runtime Environment

Environment Management
Anaconda Distribution & Python 3.12

Session Control & Navigation
Multi-page navigation via Streamlit Sidebar

MODEL PARAMETERS

Diagnostic Input Features

Heart Disease

Age & Blood Pressure: Key vital indicators.

Cholesterol Level: Serum cholesterol in mg/dl.

Resting ECG: Electrocardiographic results.

Max Heart Rate: Maximum rate achieved.

Breast Cancer

Radius Mean: Distances from center to perimeter.

Area Mean: Tumor cell size estimation.

Concavity: Severity of concave contour portions.

Symmetry: Structural cellular balance.

Parkinson's & Diabetes

Vocal Frequencies: MDVP jitter/shimmer measurements.

Glucose Levels: Plasma glucose concentration.

Insulin: 2–Hour serum insulin test.

BMI: Body mass index value.

IMPLEMENTATION NOTES

Operational Challenges & Solutions

Data Quality

Challenge: Missing metrics and clinical noise in raw data.

Solution: Automated preprocessing and imputation protocols.

Privacy & Security

Challenge: Handling sensitive patient medical history.

Solution: On-premise execution and data anonymization.

Clinical Adoption

Challenge: Hesitation towards automated black-box predictions.

Solution: Intuitive UX and clear feature importance visibility.

Conclusion & Future Vision

This platform bridges AI technology and clinical expertise to deliver rapid, accurate early diagnostics, paving the way for proactive healthcare.

Open for Questions & Discussion | Thank You for Your Attention